SABARAGAMUWA UNIVERSITY OF SRI LANKA
FACULTY OF COMPUTING
DEPARTMENT OF COMPUTING & INFORMATION SYSTEMS
BSc HONOURS DEGREE PROGRAMME IN COMPUTING AND INFORMATION SYSTEMS
IS 8101 Research Project in Information Systems

Proposal Defense Committee Review Comments

Student Name	D.K.H Dissanayake	Student Index Number	19APC3964
Research Topic	Optimizing Red/White Wine Quality Prediction Through Hybrid Ensemble Learning and Interpretable Feature Engineering.		

Comments provided:

1. Revisit the research questions and research objectives.

Research Proposal - Revised

Student Name	D.K.H Dissanayake	Student Index Number	
Research Topic	Optimizing Red/White Wine Quality Prediction Through Hybrid Ensemble Learning and Interpretable Feature Engineering.		
Supervisor(s)	Prof. B.T.G.S Kumara		

Agreement

I ……D.K.H Dissanayake……, willing to address the comments provided by the committee related to my research and will address them in the thesis. I am aware of the consequences occurs in the case of failure to address them.

…… ……05/01/2025……
Signature Date

Background

Since ancient times wine production has combined the elements of agriculture and chemistry with human evaluations. Traditional wine production has emerged as an optimal field for data science and machine learning applications during recent times. Because consumers want more wine and intense worldwide market rivalry the winemaking industry faces mounting pressure to deliver standardized product quality. The predictive method of objective quality assessment based on chemical analysis and statistical methodology provides a promising improvement to

traditional sensory evaluations that remain subjective while being expensive and time-consuming.

Research using extensive datasets such as the Red and White Wine Quality datasets obtained from Portugal's Vinho Verde region allows scientists to establish numerical relationships between wine quality sensory assessments and physical characteristics including pH and acidity as well as alcohol content and residual sugar levels. Wine quality prediction models need to demonstrate accurate performance while providing understanding that helps optimize wine production.

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The research field now explores hybrid ensembles which unite classification and regression outputs for exploiting their respective strengths. The weights assigned through dynamic ensemble approaches relate to predictive metrics and feature relevance to enhance the adaptation of heterogeneous data like wine quality datasets.

A model needs both accuracy and interpretability for becoming useful in wine production practice. The industry-wide adoption of SHAP (SHapley Additive exPlanations) along with LIME (Local Interpretable Model-agnostic Explanations) has become prevalent for explanation of complex models. The evaluation methods help reveal how distinct characteristics affect wine quality through predictions for winemakers to understand their essential properties.

Interpretability enables users to select important features and engineer them for better performance alongside lower computational requirements. The study aims to construct understandable pipelines for winemaking processes by detecting which wine parameters such as alcohol levels or pH conditions or sulfates produce maximum quality forecasts.

The proposed study develops a dual regression and classification ensemble framework that applies interpretation techniques to feature engineering and SHAP/LIME-based analyses to fill existing research gaps. The research will solve class imbalance issues by employing resampling

techniques SMOTE-Tomek while investigating domain-specific adaptive sampling approaches which give priority to alcohol and volatile acidity characteristics.

The research benefits from utilizing red and white wine datasets which bring innovative value to wine quality prediction by expanding generalizability. Even though technical improvements represent one aspect of the expected outcomes our work also presents actionable suggestions for winemakers about chemical property alterations leading to specific outcomes.

Literature Review

Traditional wine quality assessment relies on sensory evaluation by human experts, who analyze visual, olfactory, and gustatory attributes such as color clarity, aroma complexity, and balance of acidity and tannins.[1] [2]These methods, while foundational, face challenges due to subjectivity and inconsistency among tasters. For instance, Hodgson (2008) demonstrated that expert judges often assign conflicting scores to identical wines, with ratings ranging from "no award" to "gold medal" for the same sample.[3] To mitigate subjectivity, physicochemical tests emerged as objective measures, focusing on parameters like alcohol content, pH, volatile acidity, and sulfates.[4], [5] However, these tests alone cannot fully capture the nuanced sensory experience that defines consumer preference.[6] [6], [7]The disconnect between laboratory metrics and human perception underscores the need for hybrid approaches that bridge technical and sensory insights.[8]

Machine Learning (ML) has revolutionized wine quality prediction by automating the analysis of physicochemical data. Algorithms such as Random Forest (RF) and XGBoost have achieved accuracies exceeding 85–90% by identifying nonlinear relationships between features like alcohol, volatile acidity, and sulfates.[9], [10] For example, SHAP analysis in multiple studies confirmed that alcohol content (positively) and volatile acidity (negatively) are the strongest predictors of quality.[11], [12] Regression models, including Gradient Boosting Regressors (GBR), outperform traditional linear methods like Ridge Regression by capturing complex interactions between variables (e.g., sulfates enhancing the stability of phenolic compounds).[13] Deep learning architectures, such as multilayer perceptrons (MLPs), further improve predictions by modeling high-dimensional interactions, though they require larger datasets for training.[14], [15]

Wine quality datasets often exhibit severe class imbalance, with premium-quality wines (scores ≥7) representing fewer than 10% of samples.[6], [16] This skew leads models to prioritize majority classes (e.g., "average" wines), reducing sensitivity to rare but commercially critical categories. To address this, synthetic oversampling techniques like SMOTE and its variants (SMOTE-ENN, SMOTE-Tomek) generate synthetic minority-class samples, improving recall for premium wines by 15–20%.[17] [18]For instance, SMOTE-ENN combined with Extra Trees Classifier achieved 95.8% accuracy on a multiclass-imbalanced red wine dataset. However, oversampling risks introducing noise, particularly when applied to small datasets [17] (<2,000 samples), necessitating careful validation.[14]

Ensemble methods combine multiple base models (e.g., RF, XGBoost, SVMs) to enhance predictive robustness. Stacking ensembles, which use a meta-model to aggregate predictions from diverse algorithms, achieved 87–90% accuracy in recent studies, outperforming individual models by 3–5%.[14] Hybrid frameworks that integrate classification and regression outputs further refine predictions. For example, a meta-ensemble weighting SVM (classification) and XGBoost (regression) improved accuracy to 91.15% by leveraging both discrete quality categories and continuous score ranges.[19] Notably, dynamic ensembles that adapt weights based on feature subsets show promise for handling region-specific terroir effects.[20]

The "black-box" nature of complex ML models poses challenges for winemakers seeking actionable insights. SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) address this by quantifying feature contributions. For instance, SHAP values revealed that alcohol levels >11.5% ABV increase the likelihood of a wine being classified as premium by 22%, while volatile acidity >0.6 g/L reduces quality scores by 0.38 points on average.[21] [22]Such interpretability enables targeted interventions, such as adjusting fermentation times to optimize alcohol content or using sulfur dioxide to control volatile acidity.[23], [24]

Gaps in Existing Research – Detailed Study

Research mainly targets physicochemical elements (alcohol, volatile acidity) without including sensory attributes (fruity, oaky) or consumer opinions. Despite being ranked as the leading quality indicator by SHAP analysis alcohol remains unexplained regarding its connection

between chemical metrics and sensory quality features such as aroma complexity and mouthfeel perception.[25] [26]The Infowine study proved that sensory interactions between odor and astringency affect sensory evaluation results yet most machine learning algorithms analyze these variables independently.[27] The oversight of integrated sensory information makes it difficult for these systems to match expert evaluation methods that depend on total sensory integration.[28] Modern prediction models choose laboratory results above real-world consumer behavior because consumer preferences differ from expert ratings because of personal and cultural influences.[29]

The application of models trained on regions such as Portuguese Vinho Verde results in underperformance on wines produced in different terroir areas. The Kaggle Wine Quality data focuses exclusively on Portuguese wines so its feature ranking becomes distorted.[30] [31]The stabilizing role of sulfates in Vinho Verde wines appears unimportant for Napa Valley Cabernets because of distinct viticultural practices in the region. The MDPI research about New Zealand Pinot Noir wines confirmed that alcohol no longer has predictive capabilities in regions with cool climate conditions when acidity becomes the leading factor for quality perception by assessors. The usage of federated learning frameworks in enology research remains restricted since they provide collaboration options for regional model training.[32]

Wine quality treatments in most ML models function as unchanging elements even though chemical transformations happen throughout wine aging processes. The Toro D.O. study discovered that phenolic compounds and sulfite levels follow unique nonlinear patterns of transformation during wine aging while failure happens when models attempt to determine quality in wines less than four months old. The ANN model demonstrated less than 50% accuracy for wines that had reached one month in age because early chemical compositions were not distinct enough. Present datasets currently fail to monitor sensory attributes such as tobacco and leather through time since these attributes develop beyond bottling. The insufficient data collection leads to compromised applications in cellar management systems and vintage prediction methods.[33]

Research has not yet established climate change models that analyze wine quality predictors. The ongoing increase in temperatures disrupts sugar-acid relations along with phenolic development even though existing Machine Learning models operate under static growing

environment assumptions. The alcohol percentage in Portuguese wines rose 1.5% ABV since the last decade due to climate change though the data lacks historical information about wine age.

NMR and HRMS devices monitor product quality through non-invasive means yet their basic connection to ML systems remains an issue. The MDPI research utilizing digital sensors produced vintage classification outcomes with an 89% success rate but current models mainly operate from fixed laboratory measurement data. The present lack of developed computational frameworks hinders the processing of inputs from IoT-enabled vineyards that gather real-time measurements such as soil moisture and temperature.[34]

Vertical federated learning (VFL) frameworks promise to enhance model robustness while preserving data privacy, but their application in wine research is nascent. Winemakers hesitate to share sensitive production data (e.g., yeast strains, barrel types), creating bottlenecks for collaborative AI. Additionally, models trained on biased datasets risk perpetuating inequities, such as undervaluing indigenous winemaking techniques.

The successful elimination of these knowledge gaps needs collaborative work between technical specialists from data science and enology fields along with members of the wine industry and other stakeholders. The main efforts should involve assembling diverse multimodal datasets which include temporal and regional elements and establishing federated learning frameworks to protect privacy and designing hybrid models which blend interpretability with accuracy. The achievement of predictive wine quality evaluation requires the integration of ML capabilities across all relevant domains.[35]

Problem Statement and Research Questions

Traditional methods for assessing wine quality, such as expert tasting, are subjective, time-consuming, and not scalable for large-scale production. While machine learning models offer a more objective and efficient alternative by leveraging physicochemical attributes, existing approaches face challenges including limited feature relevance analysis, issues with class imbalance, and suboptimal predictive accuracy. There is a need to systematically identify key predictive features, address data imbalance, and explore advanced and hybrid machine learning models to enhance the reliability and interpretability of wine quality predictions.

Research Questions:

RQ1: What are the most influential physicochemical parameters for predicting the quality of red and white wines based on hybrid ensemble learning models?

RQ2: How do different data balancing techniques impact the performance and robustness of machine learning models in predicting imbalanced wine quality classes?

RQ3: Can hybrid ensemble models combining classification and regression approaches improve accuracy and interpretability compared to single-machine learning models in wine quality prediction?

RQ4: How can interpretable machine learning methods (e.g., SHAP, LIME) be utilized to provide actionable insights into feature contributions for practical wine quality assessment?

Research Objectives and Goals

Main Objective

To develop robust and interpretable hybrid ensemble machine learning models that enhance the accuracy and transparency of red and white wine quality predictions, addressing data imbalance and enabling actionable insights for winemakers.

Sub-Objectives

1. **Feature Analysis:** Identify key physicochemical features influencing red and white wine quality through statistical analysis and feature importance derived from ensemble models.
2. **Class Imbalance Adjustment:** Investigate and compare data balancing methods (such as SMOTE and its variants) to mitigate imbalanced quality classes and improve model fairness across categories.
3. **Model Development:** Design, implement, and optimize hybrid ensemble models that combine classification and regression for improved predictive performance.
4. **Interpretability:** Apply SHAP and LIME to explain model predictions and uncover feature interactions for practical insights relevant to winemaking processes.

5. **Evaluation and Generalization:** Validate model performance using cross-validation and, if possible, across red and white wine datasets to ensure generalizability and real-world applicability.

Figure 1 illustrates the how the research questions (RQs) align with the main and sub objectives of this research.

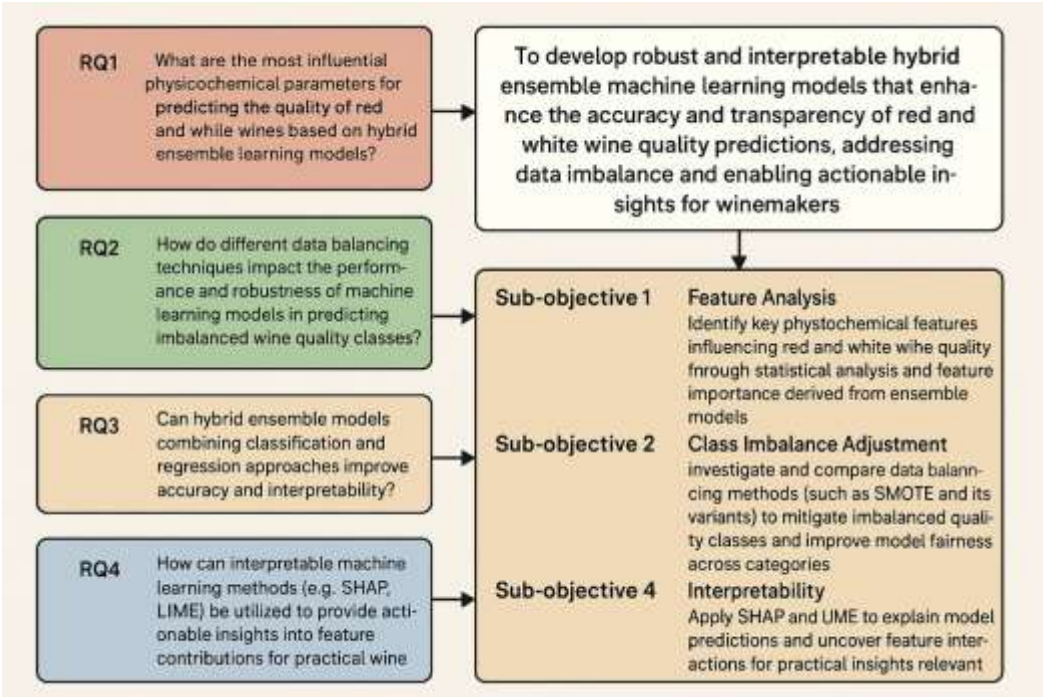

Figure 1

Proposed Research Method

1. **Data Acquisition and Exploration**
   - Obtain the red wine quality dataset from Kaggle, which includes physicochemical properties and quality ratings.
   - Perform exploratory data analysis (EDA) to understand the distribution of features and identify potential outliers, missing values, and class imbalance.
2. **Data Preprocessing**
   - Missing Values & Outliers: Address missing data and outliers using imputation and statistical techniques.

- Feature Scaling: Apply normalization or standardization to ensure all features contribute equally to model training.
- Class Imbalance Handling:
  - Analyze the distribution of wine quality classes.
  - Apply oversampling (e.g., SMOTE, SMOTE-ENN) and/or undersampling techniques to balance the dataset, as imbalanced classes can bias the model towards majority classes.
  - Compare the effectiveness of different resampling methods.

3. Feature Selection and Engineering

- Use statistical correlation (e.g., Pearson coefficient) and model-based feature importance (e.g., Random Forest, XGBoost, SHAP values) to identify and select the most influential physicochemical properties for wine quality prediction
- Remove redundant or highly collinear features to improve model interpretability and performance.

4. Model Development

Baseline Models: Train and evaluate traditional machine learning models such as:

- Support Vector Machine (SVM)
- Naïve Bayes (NB)
- Ridge Regression
- Artificial Neural Network (ANN)
- Random Forest (RF)
- Gradient Boosting Regressor (GBR)

Ensemble and Hybrid Models:

- Develop ensemble models (e.g., RF, XGBoost, Extra Trees, Voting Classifier) and hybrid approaches that combine multiple algorithms to improve predictive accuracy and robustness.
- Fine-tune model hyperparameters using grid search or metaheuristic optimization.

Model Evaluation:

- Use cross-validation and metrics such as accuracy, F1-score, recall, precision, ROC-AUC, and mean squared error (for regression tasks) to assess model performance, with a focus on minority classes.

5. **Model Interpretability**
   - Apply interpretable machine learning techniques (e.g., SHAP, LIME) to the best-performing models to understand feature contributions and provide actionable insights for winemakers.
   - Visualize feature importance and model decision boundaries.
6. **Comparative Analysis**
   - Compare the performance of individual, ensemble, and hybrid models on balanced and imbalanced data.
   - Assess the impact of feature selection and data balancing on model accuracy and interpretability.
7. **Validation and Practical Application**
   - Validate the final model using a hold-out test set or cross-dataset validation (if available).
   - Discuss the practical implications of the findings for the wine industry, including how the models and feature insights can inform wine production and quality control.
8. **Documentation and Reporting**
   - Document all processes, results, and code for reproducibility.
     - Present findings in a clear and accessible format for both technical and non-technical stakeholders.

Figure 2 illustrates the proposed methodology for developing the classification model.

Figure 2

Expected Outcome

1. **Identification of Key Predictive Features**
   - A clear ranking of the most influential physicochemical properties (such as alcohol content, volatile acidity, sulfates, and pH) that significantly impact wine quality predictions.
   - Actionable insights for winemakers on which chemical parameters to monitor and optimize during production.
2. **Improved Predictive Accuracy**
   - Machine learning models (especially ensemble and hybrid approaches) that outperform traditional single-algorithm methods in predicting wine quality, with higher accuracy, F1-score, and robustness-particularly for minority (premium) quality classes.
   - Demonstrated effectiveness of class imbalance handling techniques (e.g., SMOTE, SMOTE-ENN) in improving model sensitivity to underrepresented quality categories.
3. **Enhanced Model Interpretability**
   - Interpretable machine learning models using SHAP or LIME, providing transparent explanations of how specific features influence wine quality predictions.
   - Visualizations and reports that help stakeholders understand and trust the model's decision-making process.
4. **Practical Recommendations for the Wine Industry**
   - Guidelines for applying the developed models in real-world wine quality assessment and quality control processes.
   - Recommendations for data collection and feature monitoring to further improve future predictive models.
5. **Reproducible Research Artifacts**

- Well-documented code, data processing pipelines, and model evaluation scripts, enabling reproducibility and future extension of the research.

Research Plan and Time Table

Table 1:Research plan Time Table

	2025								2026	
Activity	May	June	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb
Selection a topic										
Literature Review										
Find optimal method										
Develop the basic models										
Publishing of the research										

Proposed Budget

The Internet and publications costs are the expenses for the research.

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Any Other Relevant Information